

Corporate Finance

Part 1. Applying deep learning to structural corporate finance model

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Abstract

This project applies deep learning techniques to solve a structural dynamic corporate finance model with investment and risky debt. The firm's problem is formulated as a discrete-time dynamic programming model with capital adjustment costs, stochastic productivity, and endogenous default. The basic investment model and its extension with risky debt [9] are solved using the all-in-one Bellman residual method of Maliar et al. [8], with value and policy functions approximated by neural networks and trained in TensorFlow. The results demonstrate stable convergence and economically plausible dynamics, indicating the effectiveness of deep learning in solving dynamic corporate finance models.

1 Introduction

Dynamic corporate finance models study firm decisions that evolve over time under uncertainty. Investment, financing, and default choices interact dynamically and cannot be understood using static frameworks. Firms face persistent productivity shocks, capital adjustment frictions, and intertemporal trade-offs between current payouts and future value. These features require a dynamic optimization framework in which firm value is defined recursively.

This project studies a discrete-time investment model drawn from the structural corporate finance literature. The baseline framework follows the discrete-time dynamic investment model developed by Strebulaev and Whited [9]. The firm chooses an investment each period to maximize shareholder value, taking into account adjustment costs and stochastic productivity. The model is formulated as an infinite-horizon dynamic programming problem, with firm value characterized by a Bellman equation.

1.1 Basic Investment Model

We consider a firm that operates in discrete time with an infinite planning horizon. The firm is owned by risk-neutral shareholders and seeks to maximize the expected present value of payouts to equity holders. Time is indexed by $t = 0, 1, 2, \dots$

State Variables At the beginning of period t , the firm is characterized by two state variables:

1. k_t , the capital stock available for production,
2. z_t , an exogenous productivity shock.

The productivity shock follows a Markov process. In particular, the logarithm of productivity evolves according to an autoregressive process, AR(1):

$$\ln z_{t+1} = \rho \ln z_t + \varepsilon_{t+1}, \quad (1)$$

where $|\rho| < 1$ and ε_{t+1} is an independent and identically distributed innovation with zero mean.

Production and Profits Output is produced using capital according to a profit function $\pi(k_t, z_t)$. The profit function satisfies the standard regularity conditions:

$$\pi_k(k_t, z_t) > 0, \quad \pi_{kk}(k_t, z_t) \leq 0, \quad \pi_z(k_t, z_t) > 0.$$

Capital Accumulation Investment undertaken in period t , denoted by I_t , augments the capital stock subject to depreciation:

$$k_{t+1} = (1 - \delta)k_t + I_t, \quad (2)$$

where $\delta \in (0, 1)$ is the constant depreciation rate of capital.

Adjustment Costs Investment is costly due to installation, organizational, and adjustment frictions. These costs are captured by an adjustment cost function $\psi(I_t, k_t)$. The adjustment cost function is increasing in I_t and weakly decreasing in k_t .

Cash Flow to Shareholders The firm's net payout to shareholders in period t , denoted by e_t , is determined by:

$$e_t = \pi(k_t, z_t) - I_t - \psi(I_t, k_t). \quad (3)$$

Positive values of e_t correspond to dividend payments to shareholders. Negative values represent equity injections from shareholders. In the basic model, equity financing is frictionless, so negative payouts are allowed without penalty.

Firm Value and Dynamic Optimization Let $V(k_t, z_t)$ denote the value of the firm at the beginning of period t . The firm chooses future capital stocks to maximize the expected discounted sum of payouts to shareholders:

$$V(k_t, z_t) = \max_{\{k_{t+j}\}_{j=1}^{\infty}} \mathbb{E}_t \left[\sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j e_{t+j} \right], \quad (4)$$

where r is the constant risk-free interest rate and $\mathbb{E}_t[\cdot]$ denotes expectations conditional on information available at time t .

Using the principle of optimality, this infinite-horizon problem can be written recursively. Substituting the capital accumulation equation and expressing investment as

$$I_t = k_{t+1} - (1 - \delta)k_t, \quad (5)$$

the value function satisfies the Bellman equation:

$$V(k_t, z_t) = \max_{k_{t+1}} \left\{ \pi(k_t, z_t) - \psi(k_{t+1} - (1 - \delta)k_t, k_t) - (k_{t+1} - (1 - \delta)k_t) + \frac{1}{1 + r} \mathbb{E}_t [V(k_{t+1}, z_{t+1})] \right\}. \quad (6)$$

This Bellman equation characterizes firm value recursively as the sum of current net cash flows and the discounted expected continuation value.

1.2 Risky Debt Model

The basic investment model assumes frictionless equity financing and rules out default by construction. In reality, firms rely heavily on debt financing, and adverse economic shocks may render them unable or unwilling to meet their debt obligations. We now extend the baseline investment model to incorporate risky debt and endogenous default. In this model, two new economic elements are introduced. First, debt is risky because repayment is contingent on the firm's future value. Second, equity holders have limited liability and may optimally choose to default when continuation is unattractive.

New Elements Relative to the Basic Model Relative to the basic model, two modifications are introduced.

1. First, the firm now carries outstanding one-period debt. Let b_t denote debt due at the end of period t . The firm chooses next-period debt b_{t+1} alongside next-period capital.
2. Second, debt is priced at a risky interest rate. Let $\tilde{r}_t$ denote the interest rate on debt issued in period t .

Risky Debt Pricing and Lender Zero-Profit Condition Lenders are competitive and risk-neutral. They price debt so that the expected return equals the risk-free rate. Because default is possible, lenders may not receive the promised repayment in all future states.

If the firm does not default, lenders receive the full promised repayment:

$$b_{t+1}(1 + \tilde{r}_t). \quad (7)$$

If the firm defaults, lenders recover only a fraction of the firm's value. Recovery is given by:

$$R(k_{t+1}, z_{t+1}) = (1 - \alpha) [(1 - \tau)\pi(k_{t+1}, z_{t+1}) + (1 - \delta)k_{t+1}], \quad (8)$$

where $\alpha \in (0, 1)$ captures deadweight default costs.

Let $\mathcal{D}_{t+1}$ denote the set of states in which default occurs, and let $\mathcal{S}_{t+1}$ denote the set of solvent states. The lender's zero-profit condition requires that the expected payoff on debt equal

the risk-free return:

$$b_{t+1}(1+r) = \mathbb{E}_t [\mathbf{1}_{\mathcal{S}_{t+1}} b_{t+1}(1+\tilde{r}_t) + \mathbf{1}_{\mathcal{D}_{t+1}} R(k_{t+1}, z_{t+1})]. \quad (9)$$

Solving for the risky interest rate gives:

$$1 + \tilde{r}_t = \frac{1 + r - \mathbb{E}_t \left[\mathbf{1}_{\mathcal{D}_{t+1}} \frac{R(k_{t+1}, z_{t+1})}{b_{t+1}} \right]}{\Pr_t(\mathcal{S}_{t+1})}. \quad (10)$$

Cash Flow to Equity with Risky Debt The firm's sources and uses of funds identity is modified to account for risky debt issuance and repayment. After-tax operating profits are given by:

$$(1 - \tau)\pi(k_t, z_t).$$

Investment and adjustment costs reduce cash flows by:

$$I_t + \psi(I_t, k_t).$$

Issuing new debt b_{t+1} generates resources equal to the discounted value of promised repayment:

$$\frac{b_{t+1}}{1 + \tilde{r}_t}.$$

Interest payments generate a tax shield. For tractability, Strebulaev [9] assumes that the present value of the interest tax deduction is realized immediately:

$$\frac{\tau \tilde{r}_t b_{t+1}}{(1 + \tilde{r}_t)(1 + r)}.$$

Finally, existing debt b_t must be repaid in full if default does not occur.

Combining all components, net payouts to equity holders are:

$$\begin{aligned} e_t &= (1 - \tau)\pi(k_t, z_t) - I_t - \psi(I_t, k_t) \\ &\quad + \frac{b_{t+1}}{1 + \tilde{r}_t} + \frac{\tau \tilde{r}_t b_{t+1}}{(1 + \tilde{r}_t)(1 + r)} - b_t. \end{aligned} \quad (11)$$

Bellman Equation with Endogenous Default Let $V(k_t, b_t, z_t)$ denote equity value at the beginning of period t . Conditional on continuation, equity holders choose next-period capital and debt to maximise shareholder value.

The continuation value equals current net payouts plus the discounted expected value of future equity:

$$e_t + \frac{1}{1 + r} \mathbb{E}_t [V(k_{t+1}, b_{t+1}, z_{t+1})]. \quad (12)$$

Because default is an option, equity value is the maximum of zero and the continuation value. The Bellman equation is therefore:

$$V(k_t, b_t, z_t) = \max \left\{ 0, \max_{k_{t+1}, b_{t+1}} \left[e_t + \frac{1}{1 + r} \mathbb{E}_t [V(k_{t+1}, b_{t+1}, z_{t+1})] \right] \right\}. \quad (13)$$

2 Literature Review

Dynamic corporate finance problems are typically formulated as stochastic dynamic programming models, requiring numerical solution methods to handle optimization under uncertainty. Early work relied on value function iteration on discretized state spaces [7]. This approach is conceptually straightforward and can accommodate non-convexity and discrete choices, making it suitable for early dynamic corporate finance models with default [1]. However, its computational cost grows exponentially with the number of state variables, leading to the well-known curse of dimensionality.

To improve efficiency, researchers developed projection and Euler-equation-based methods, which approximate policy functions using parametric basis functions and enforce first-order conditions at selected points [7]. These methods are effective for smooth problems and are widely used in macroeconomic applications. Their reliance on differentiability, however, limits their applicability in models with adjustment costs, borrowing constraints, or default, where Euler equations may fail to hold globally [9]. A further step forward was made with simulation-based methods, such as the parameterized expectations algorithm [4], which approximates conditional expectations using simulated data rather than fixed grids. By focusing computation on the ergodic region of the state space, these methods partially mitigate dimensionality concerns. Nevertheless, they can suffer from numerical instability and sensitivity to functional specification, particularly in complex or non-smooth models [3].

More recently, machine learning approaches, especially neural networks, have been explored as flexible function approximators for value and policy functions. Early applications used neural networks primarily as interpolation tools [5], while later work integrated them more directly into solution algorithms [2, 6]. Neural networks are well-suited to capturing nonlinearities and high-dimensional interactions, but their effectiveness depends critically on how they are trained to satisfy the underlying economic equilibrium conditions.

In this context, residual-based deep learning methods represent a significant advance. Maliar et al. [8] propose a unified all-in-one (AiO) framework that reformulates dynamic economic models as residual minimization problems. The method enforces the Bellman equation and, where appropriate, Euler equations directly using simulated data and stochastic gradient optimization. By combining Monte Carlo simulation with flexible neural network approximations, the AiO method avoids fixed grids, mitigates the curse of dimensionality, and can handle non-smooth features. These developments motivate the use of residual-based deep learning methods for solving the models considered in this project.

3 Methodology

This section explains how the economic model derived in Section 1 is solved numerically. We explain how the Bellman equations are transformed into a system of residual equations and solved using the all-in-one (AiO) Bellman-residual approach proposed by Maliar et al. [8].

3.1 Basic Investment Model

The presence of the maximization operator makes the Bellman equation non-differentiable in its raw form. To construct a system of equilibrium conditions suitable for numerical solution, we exploit the fact that interior solutions must satisfy first-order optimality conditions.

3.1.1 First-Order Condition for Optimal Investment

Define the Bellman objective function inside the maximization as:

$$\mathcal{G}(k_t, z_t, k_{t+1}) = \pi(k_t, z_t) - \psi(I_t, k_t) - I_t + \beta \mathbb{E}_t [V(k_{t+1}, z_{t+1})], \quad (14)$$

with $I_t = k_{t+1} - (1 - \delta)k_t$.

For an interior optimum, the necessary first-order condition with respect to k_{t+1} is:

$$\frac{\partial \mathcal{G}(k_t, z_t, k_{t+1})}{\partial k_{t+1}} = 0. \quad (15)$$

The profit function does not depend on next-period capital, so:

$$\frac{\partial \pi(k_t, z_t)}{\partial k_{t+1}} = 0.$$

Investment enters linearly through I_t , so:

$$\frac{\partial I_t}{\partial k_{t+1}} = 1.$$

The derivative of adjustment costs is:

$$\frac{\partial \psi(I_t, k_t)}{\partial k_{t+1}} = \psi_I(I_t, k_t) \frac{\partial I_t}{\partial k_{t+1}} = \psi_I(I_t, k_t).$$

Finally, the derivative of the continuation value is:

$$\frac{\partial}{\partial k_{t+1}} (\beta \mathbb{E}_t [V(k_{t+1}, z_{t+1})]) = \beta \mathbb{E}_t [V_k(k_{t+1}, z_{t+1})].$$

Combining all terms yields the Euler equation:

$$\psi_I(I_t, k_t) + 1 = \beta \mathbb{E}_t [V_k(k_{t+1}, z_{t+1})]. \quad (16)$$

3.1.2 Equilibrium Conditions and Residual Representation

In equilibrium, two conditions must hold simultaneously at every point in the state space. First, the value function must satisfy the Bellman equation. Define the Bellman residual as:

$$R_V(k_t, z_t) = V(k_t, z_t) - [\pi(k_t, z_t) - \psi(I_t, k_t) - I_t + \beta \mathbb{E}_t [V(k_{t+1}, z_{t+1})]]. \quad (17)$$

Second, the first-order condition must hold. Define the Euler residual as:

$$R_{\text{FOC}}(k_t, z_t) = \psi_I(I_t, k_t) + 1 - \beta \mathbb{E}_t [V_k(k_{t+1}, z_{t+1})]. \quad (18)$$

A perfect solution to the model satisfies:

$$R_V(k_t, z_t) = 0, \quad R_{\text{FOC}}(k_t, z_t) = 0 \quad (19)$$

for all (k_t, z_t) .

The unknown objects are the value function $V(\cdot)$ and the policy function $h(\cdot)$. Traditional numerical methods approximate these functions using grids and value function iteration. This approach becomes computationally expensive in higher dimensions.

3.1.3 Maliar All-in-One (AiO) Method

The AiO method proposed by Maliar et al. [8] replaces value function iteration with direct minimization of equilibrium residuals. The key idea is to approximate the value function and the policy function jointly using neural networks. Formally, the loss function is defined as:

$$\mathcal{L} = \mathbb{E} \left[R_V(k_t, z_t)^2 + \lambda R_{\text{FOC}}(k_t, z_t)^2 \right], \quad (20)$$

where $\lambda > 0$ is a weighting parameter.

A practical difficulty with directly minimizing squared residuals is that the expectation operator appears inside a nonlinear transformation. Maliar et al. [8] resolve this issue using an all-in-one trick based on independent simulation draws. Let ε and ε' denote two independent draws from the same shock distribution. Then the square of the expectation can be written as:

$$(\mathbb{E}[R(x, \varepsilon)])^2 = \mathbb{E} [R(x, \varepsilon) R(x, \varepsilon')]. \quad (21)$$

Applying this idea to both equilibrium conditions yields the AiO loss:

$$\mathcal{L} = \mathbb{E} [R_V(k_t, z_t, \varepsilon) R_V(k_t, z_t, \varepsilon') + \lambda R_{\text{FOC}}(k_t, z_t, \varepsilon) R_{\text{FOC}}(k_t, z_t, \varepsilon')], \quad (22)$$

where ε and ε' are independent conditional on the current state (k_t, z_t) .

3.2 Risky Debt Model

The continuation value is

$$C(k_t, b_t, z_t) = e(k_t, b_t, z_t) + \beta \mathbb{E}_t [V(k_{t+1}, b_{t+1}, z_{t+1})]. \quad (23)$$

Limited liability implies the inequality constraint

$$V(k_t, b_t, z_t) \geq 0, \quad (24)$$

with default occurring whenever $C(k_t, b_t, z_t) \leq 0$.

In the basic model, the value function is smooth, and the Bellman equation can be differentiated to obtain Euler equations. That approach fails here for two fundamental reasons. First, the Bellman equation contains a max operator. Second, default introduces an occasionally binding inequality constraint.

3.2.1 Bellman Residual Formulation

A correct solution must satisfy the Bellman equation pointwise. Define the Bellman residual:

$$R_V(k_t, b_t, z_t) = V(k_t, b_t, z_t) - \max \left\{ 0, e(k_t, b_t, z_t) + \beta \mathbb{E}_t[V(k_{t+1}, b_{t+1}, z_{t+1})] \right\}. \quad (25)$$

At the true solution,

$$R_V(k_t, b_t, z_t) = 0 \quad \text{for all } (k_t, b_t, z_t). \quad (26)$$

3.2.2 AiO Reformulation of the Loss Function

The AiO method uses the identity

$$(\mathbb{E}[R_V])^2 = \mathbb{E}[R_V(\varepsilon) R_V(\varepsilon')], \quad (27)$$

where ε and ε' are independent draws from the same shock distribution.

For each state (k_t, b_t, z_t) , draw two independent shocks $\varepsilon, \varepsilon'$ and define:

$$\begin{aligned} R_{V,1} &= V(k_t, b_t, z_t) - \max\{0, e_t + \beta V(k_{t+1}, b_{t+1}, z_{t+1}^{(1)})\}, \\ R_{V,2} &= V(k_t, b_t, z_t) - \max\{0, e_t + \beta V(k_{t+1}, b_{t+1}, z_{t+1}^{(2)})\}. \end{aligned} \quad (28)$$

The AiO loss function is then:

$$\mathcal{L} = \mathbb{E}[R_{V,1} \cdot R_{V,2}]. \quad (29)$$

4 Answers to the Questions

Q1. Implement the basic investment model and the Maliar method in Tensorflow.

Answer 1: The basic investment model is implemented in TensorFlow by directly translating the firm's dynamic optimization problem into a residual-based learning framework. Both the value function and the policy function are parameterized using neural networks, which take capital and productivity as inputs and return the value and optimal next-period capital, respectively.

The solution method follows the AiO approach proposed by Maliar et al. [8]. Instead of solving the model using grid-based value function iteration, the Bellman equation and the Euler equation are reformulated as residual conditions that must hold in equilibrium. These residuals are evaluated at randomly sampled states and simulated productivity shocks, allowing expectations to be approximated through Monte Carlo simulation. The problem formulation of the basic investment model into the Maliar framework is provided

in the Methodology section (see 3.1). The training procedure is described in Algorithm (1).

- Q2.** Generate a set of synthetic data to test the effectiveness of your solution method. Please define effectiveness and clearly state measures for it.

Answer 2: Synthetic data are generated by simulating time series of state variables and endogenous decisions using the trained policy function. Starting from randomly drawn initial conditions for capital and productivity, the model is simulated forward for multiple periods using the assumed stochastic process for productivity shocks. Initial productivity is sampled from the ergodic distribution implied by the AR(1) process, while future productivity is generated using the model's transition equation. This procedure produces artificial samples of capital, investment, productivity, and cash flows that are internally consistent with the model's structure.

Effectiveness of the solution method is defined as the extent to which the learned policy and value functions satisfy the equilibrium conditions of the model and generate economically meaningful outcomes. Since a closed-form analytical solution is not available, effectiveness is evaluated using indirect but theoretically grounded measures. In this study, two complementary measures are used:

- (a) **Bellman Residuals:** The Bellman residual measures the degree to which the learned value function satisfies the Bellman equation at a given state. If the Bellman equation is satisfied, the value function correctly represents the maximal attainable discounted payoff, and the associated policy function is dynamically consistent. A model is considered effective with respect to this criterion if Bellman residuals are small in magnitude, centered near zero, and stable across the relevant region of the state space.
- (b) **Lifetime Reward (LR):** The lifetime reward measures the realized discounted sum of flow payoffs generated by the learned policy when the model is simulated forward over time. It is computed by starting from an initial state, applying the policy function repeatedly, and accumulating discounted payoffs along the simulated path. A model is considered effective if the lifetime reward is stable across simulation paths and does not exhibit explosive or collapsing behavior.

- Q3.** What are the issues to consider when designing the neural network and the training algorithm?

Answer 3: The following issues are to be considered when designing the neural network and the training algorithm:

- (a) **Choice of Network Architecture:** The value function and the policy function are nonlinear functions of capital and productivity. The neural network must therefore be sufficiently flexible to approximate these nonlinear mappings. An overly simple architecture may lead to large approximation errors, while an excessively complex network may introduce overfitting and unstable training. Activation functions must

be smooth to allow reliable gradient computation, and output transformations should respect economic constraints.

- (b) **Approximation of Expectations:** The Bellman and Euler equations contain conditional expectations over future productivity shocks. The training algorithm must use independent shock draws and the AiO reformulation to obtain an unbiased objective function.
- (c) **Sampling of the State Space:** The effectiveness of the learned solution depends on how states are sampled during training. States should be drawn from economically relevant regions of the state space, ideally covering the ergodic distribution implied by the model.
- (d) **Handling of Economic Constraints:** Dynamic economic models often include inequality constraints or max operators, as in the case of default and limited liability. These features can make the value function non-differentiable. The network design and loss function must accommodate such non-smoothness.
- (e) **Numerical Stability and Convergence Diagnostics:** Value functions and residuals can vary significantly in magnitude across states. Without appropriate normalization, training may suffer from exploding or vanishing gradients. Learning rates, batch sizes, and stopping criteria must be chosen carefully.

Q4. Can the method be applied to the more sophisticated models such as the one in Section 3.6 with risky debt? If so, please show the effectiveness of the solution with synthetically generated data.

Answer 4: Yes, the Maliar method can also be applied to risky debt models. In this model, the firm’s problem is characterized by limited liability and a default option, which introduces a max operator and an inequality constraint into the Bellman equation. As a result, the value function is no longer smooth, and standard Euler equations are not valid as global equilibrium conditions.

To address this issue, the risky-debt model is formulated directly in terms of the Bellman equation for equity value, and the solution method relies exclusively on Bellman residuals rather than first-order conditions. The value function and the joint policy function for capital and debt are parameterized using neural networks, and equilibrium is enforced by minimizing Bellman residuals evaluated under independent shock draws using the AiO framework. The complete formulation is explained in the Methodology section (see 3.2). The training procedure using the TensorFlow framework is described in Algorithm (2).

5 Results

5.1 Testing Plan

Three complementary diagnostics are used for both the basic investment model and the risky-debt model. First, training convergence is evaluated by monitoring the Bellman–AiO loss during training. A decreasing and stabilizing loss indicates convergence of the learning algorithm.

Algorithm 1 AiO Training Algorithm for the Basic Investment Model

- 1: **Input:** Discount factor β , depreciation rate δ , productivity process parameters, learning rate η , batch size N , maximum iterations T .
- 2: **Objective:** Policy function $k_{t+1} = h(k_t, z_t)$ and Value function $V(k_t, z_t)$.
- 3:
- 4: **Step 1. Initialization**
- 5: Initialize parameters θ and ϕ of the policy and value networks, respectively.
- 6:
- 7: **Step 2. Iteration**
- 8: **while** $i < T$ **do**
- 9: Draw a random batch of states $\{(k^j, z^j)\}_{j=1}^N$.
- 10: For each state (k^j, z^j) , draw two independent shocks $\varepsilon_1^j, \varepsilon_2^j$.
- 11: Compute next-period capital using the policy function:

$$k'^j = h_\theta(k^j, z^j).$$

- 12: Compute next-period productivity states $z_1'^j, z_2'^j$.
- 13: Evaluate Bellman residuals under both shocks:

$$R_{V,1}^j, R_{V,2}^j.$$

- 14: Evaluate Euler-equation residuals under both shocks:

$$R_{FOC,1}^j, R_{FOC,2}^j.$$

- 15: Construct the AiO loss:

$$\mathcal{L} = \frac{1}{N} \sum_{j=1}^N \left(R_{V,1}^j R_{V,2}^j + \lambda R_{FOC,1}^j R_{FOC,2}^j \right).$$

- 16: Update parameters (θ, ϕ) .
 - 17: $i \leftarrow i + 1$.
 - 18: **end while**
 - 19:
 - 20: **Step 3. Output**
 - 21: Trained policy function h_θ and value function V_ϕ .
-

Algorithm 2 AiO Training Algorithm for the Risky Debt Model

- 1: **Given:** Discount factor β , tax rate τ , bankruptcy cost parameter, productivity process parameters, learning rate η , batch size N , maximum iterations T .
- 2: **Objective:** Policy function $h(k_t, b_t, z_t)$ and Value function $V(k_t, b_t, z_t)$.
- 3:
- 4: **Step 1. Initialization**
- 5: Initialize parameters θ and ϕ of the policy and value networks, respectively.
- 6:
- 7: **Step 2. Iteration**
- 8: **while** $i < T$ **do**
- 9: Draw a random batch of states $\{(k^j, b^j, z^j)\}_{j=1}^N$.
- 10: For each state (k^j, b^j, z^j) , draw two independent shocks $\varepsilon_1^j, \varepsilon_2^j$.
- 11: Compute next-period capital and debt using the policy function:

$$(k'^j, b'^j) = h_\theta(k^j, b^j, z^j).$$

- 12: Compute next-period productivity states $z_1'^j, z_2'^j$.
- 13: For each shock realization, compute continuation value:

$$C^j = e^j + \beta V_\phi(k'^j, b'^j, z'^j).$$

- 14: Construct Bellman targets:

$$V_{\text{target},1}^j = \max\{0, C_1^j\}, \quad V_{\text{target},2}^j = \max\{0, C_2^j\}.$$

- 15: Compute Bellman residuals:

$$R_{V,1}^j = V_\phi(k^j, b^j, z^j) - V_{\text{target},1}^j, \quad R_{V,2}^j = V_\phi(k^j, b^j, z^j) - V_{\text{target},2}^j.$$

- 16: Construct the AiO loss:

$$\mathcal{L} = \frac{1}{N} \sum_{j=1}^N R_{V,1}^j R_{V,2}^j.$$

- 17: Update parameters (θ, ϕ) .
 - 18: $i \leftarrow i + 1$.
 - 19: **end while**
 - 20:
 - 21: **Step 3. Output**
 - 22: Trained policy function h_θ and value function V_ϕ .
-

Second, out-of-sample Bellman residuals are computed on synthetically generated states not used during training. This test evaluates whether the learned value function approximately satisfies the Bellman equation across the relevant state space. Third, dynamic simulations are conducted using the learned policy functions to generate synthetic time paths. From these simulations, cumulative discounted lifetime rewards are computed to assess the stability and economic plausibility of the induced dynamics.

5.2 Results for the Basic Investment Model

Figure (1) reports the results for the basic investment model. Panel (a) shows that the Bellman-AiO loss decreases steadily and converges, indicating stable training and improved satisfaction of the equilibrium conditions. Panel (b) presents the distribution of Bellman residuals evaluated on out-of-sample synthetic data. The residuals are concentrated near zero, indicating that the learned value function approximately satisfies the Bellman equation across the state space. Panel (c) displays the cumulative discounted lifetime reward along a simulated path. The smooth and stabilizing trajectory confirms that the learned policy generates economically plausible and stable long-run behavior.

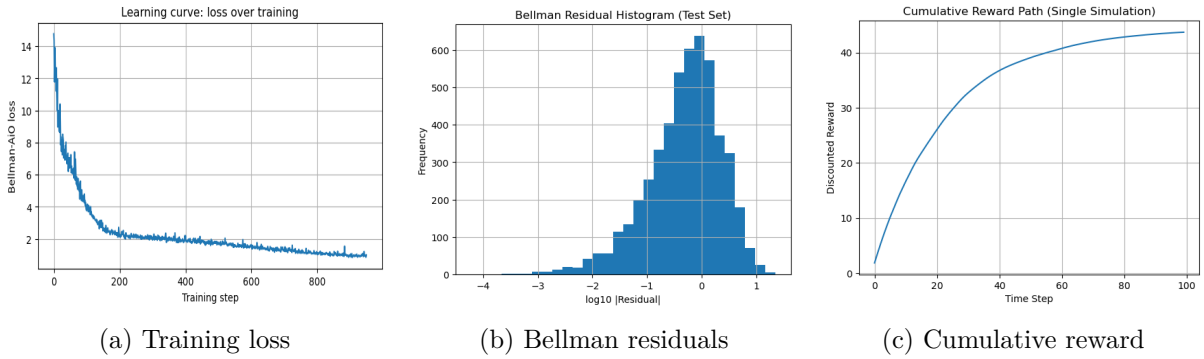


Figure 1: Training and out-of-sample diagnostics for the basic investment model.

5.3 Results for the Risky Debt Model

Figure (2) presents the corresponding results for the risky-debt model. Panel (a) shows a steadily decreasing Bellman-residual-based loss, demonstrating convergence despite the presence of non-smoothness due to default and limited liability. Panel (b) reports the distribution of Bellman residuals on out-of-sample synthetic data, with residuals centered near zero, indicating approximate satisfaction of the Bellman equation. Panel (c) shows the cumulative discounted lifetime reward from a simulated path under the learned joint policy for capital and debt. The stable trajectory indicates that the learned policy produces economically meaningful dynamics without explosive behavior.

The combined evidence from training convergence, small out-of-sample Bellman residuals, and stable lifetime reward simulations supports the correctness and validity of the implementation. These results indicate that the TensorFlow implementation of the Maliar all-in-one method successfully solves both the basic investment model and the risky-debt model with endogenous default.

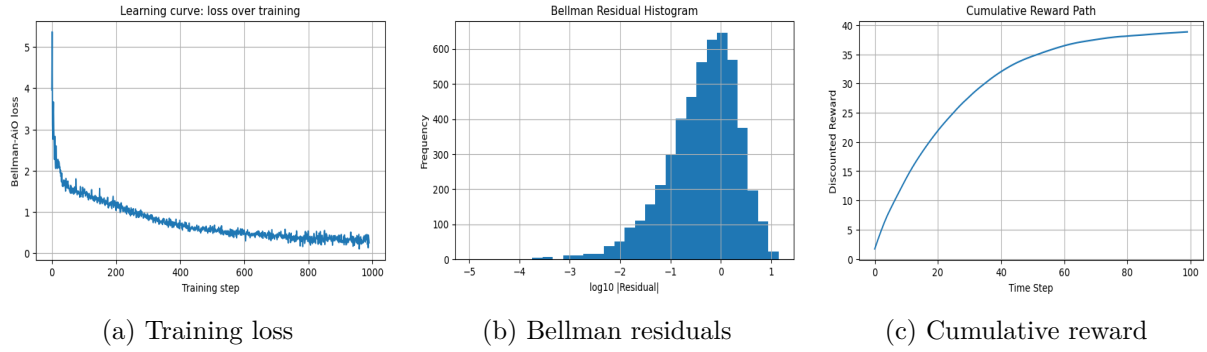


Figure 2: Training and out-of-sample diagnostics for the risky debt model.

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